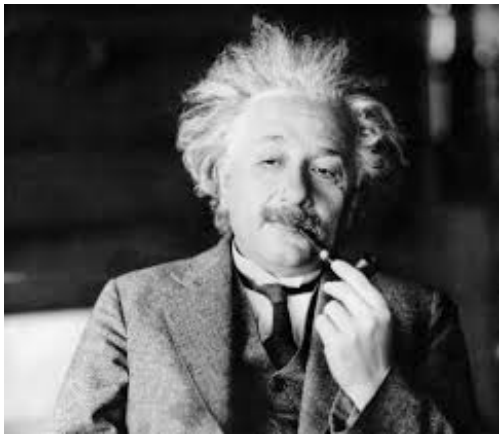
James Clerk Maxwell (13 June 1831 – 5 November 1879) was a Scottish mathematician

Einstein (14 March 1879 - 18 April 1955) was a German-born American Theoretical Physicist



The God said,

$$\vec{\nabla} \circ \vec{E} = \frac{\rho}{\varepsilon_0}$$

$$\vec{\nabla} \circ \vec{B} = 0$$

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\vec{\nabla} \times \vec{B} = \mu_0 (\vec{J} + \varepsilon_0 \frac{\partial \vec{E}}{\partial t})$$

And there was Light !

The first equation is known as differential from of Gauss's law in electrostatics for a free space.

$$\vec{\nabla} \circ \vec{E} = \frac{\rho}{\varepsilon_0}$$

ε_0 = 8.85×10^{-12} F·m^{-1}

The second equation is knows as Gauass's law in magnetostatics, and it implies the total magnetic flux taken over any closed surface is always zero as for any closed surface as many magnetic flux enters the surface from one side same amount of flux emanates from the other side.

$$\vec{\nabla} \circ \vec{B} = 0$$

The third law is the differential form of Faraday's law of electromagnetic induction. This represents electromotive force around a closed path is equal to the negative rate of change of magnetic flux with the closed path.

$$\vec{\nabla} \times \vec{E} = -u_0 \frac{\partial \vec{H}}{\partial t}$$

The fourth law is the Ampere's law, which states the circulating magnetic fields are produced by changing electric fields and by electric currents.

$$\vec{\nabla} \times \vec{H} = \vec{J} + \frac{\partial D}{\partial t} \quad \vdots \quad \vec{B} = \mu_0 \vec{H} \quad \vdots \quad \vec{D} = \varepsilon_0 \vec{E}$$

$$\vec{\nabla} \times \vec{B} = \mu_0 \left(\vec{J} + \varepsilon_0 \frac{\partial \vec{E}}{\partial t} \right) \quad \vdots \quad \mu_0 = 12.57 \times 10^{-7} \text{ H/m}$$

Taking WAVE EQUATIONS for electric and magnetic fields in vacuum, we get

$$V^2 = \frac{1}{\varepsilon_0 \mu_0} \quad \therefore \quad V = \frac{1}{\sqrt{\varepsilon_0 \mu_0}}$$

C=299,792,458 m/s

Introduction:

Before the 20th century, two branches of physics **1) Mechanics & 2) Electromagnetism** had developed quite independently of each other and it seemed like **there is no connection between the two.**
NLM couldn't quite help with objects moving with high velocities but these were so perfect for objects moving with lower speeds. Most of the failures of NLM have been overcome by Einstein's Theories about relativity.

Newtonian Relativity: Newtonian mechanics deals with speeds incomparable with the speed of light. According to it, the three fundamental concepts of physics which are space, time and mass are absolute and invariant which means these are not affected by any motion of the body.

Galilean Transformation:
If we observe an event from two different frame of reference, then a set of coordination of the event in one frame of reference are different from the other. The equation relating the two sets of coordinates of the event in the two frames of reference are called transformation equations. Galileo formulated the equations which relate the motions observed by two observers in two different inertial frames. These equation are called Galilean transformation.

Example:

{ This implies that the form of equations representing Newton's second law of motion has not changed under Galilean transformation. Hence, we say that **Newton's law is invariant under Galilean transformation.** }

Einstein in Context: 1905, Annus Mirabilis
1) On a Heuristic Point of View Concerning the Production and Transformation of Light
2) On the Movement of Small Particles Suspended in Stationary Liquids Required by the Molecular-Kinetic Theory of Heat
3) **On the Electrodynamics of Moving Bodies External**
4) Does the Inertia of a Body Depend Upon its Energy Content?

The Postulates: Ugh, it's physics.

2)Principle of Light Constancy: The **speed of light is constant in vacuum** in all inertial reference frames (i.e. it is independent from the movement of the inertial reference frame or the observer in it).

1)The Relativity Principle: A fundamental law of physics that states that **gravitational and inertial forces are of a similar nature and often indistinguishable**.

Inertial Frame: Either at rest or constant velocity

Concept of light year and synchronization of clocks and the light clock to find out gamma (γ)

A light-year is **a measurement of distance and not time**. A light-year is the distance a beam of light travels in a single Earth year, which equates to approximately 9.7 trillion kilometres.

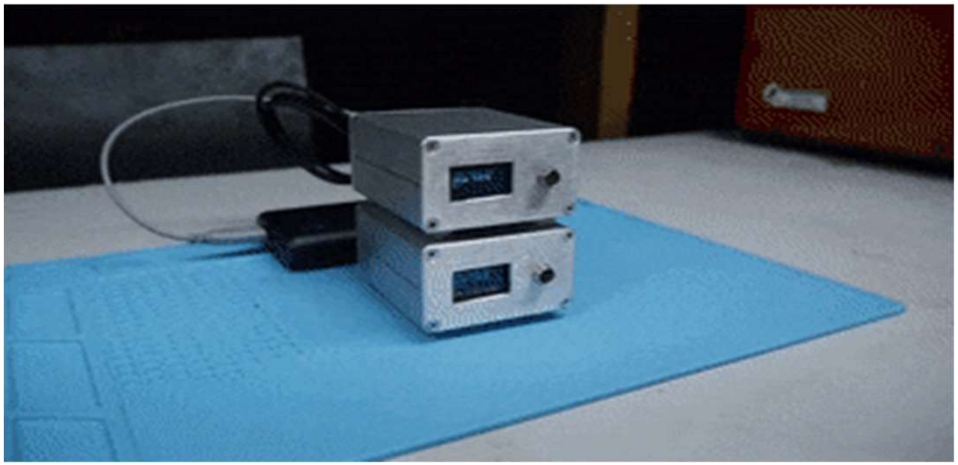
For a moment let us suppose that there-

Paradoxes to Pounder along with Real Life Examples :

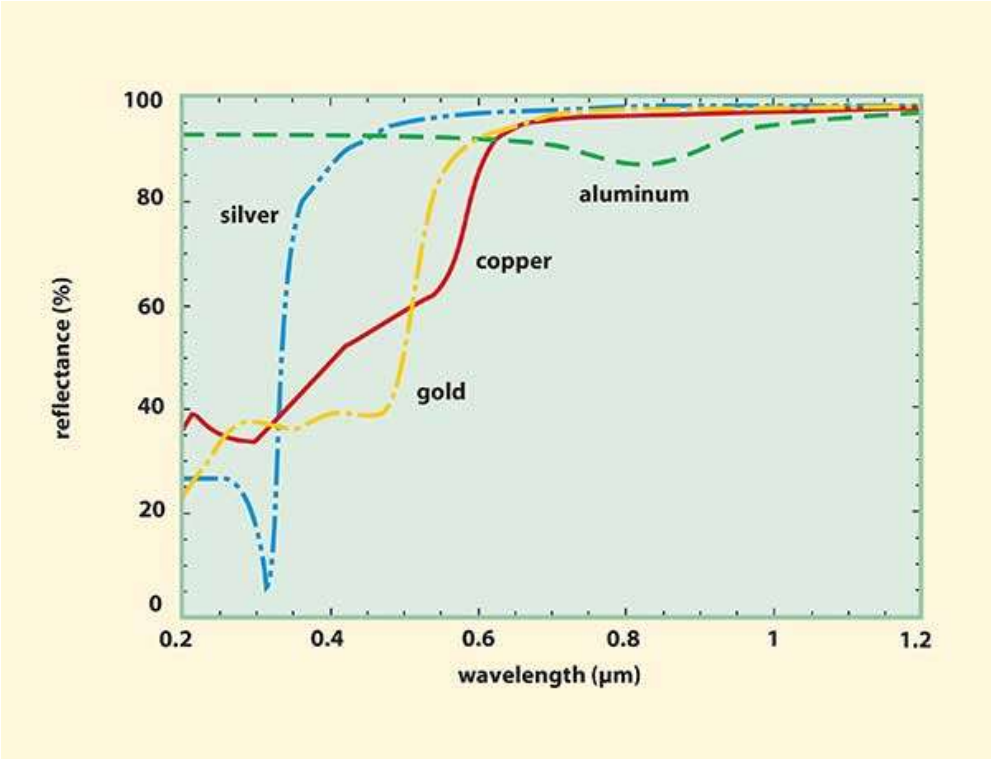
1) The Barn Paradox-

V=0.6c (1.25)

2) The Muon Paradox- V=0.998



3) GOLD -

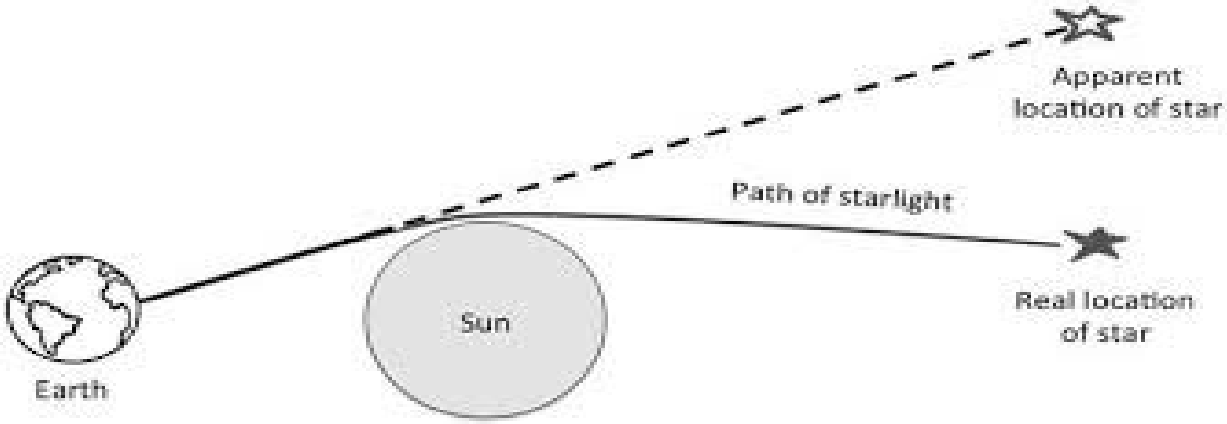


Group→ / ↓Period	1	2	3		4	5	6	7	8	9	10	11	12
1	1 H												
2	3 Li	4 Be											
3	11 Na	12 Mg											
4	19 K	20 Ca	21 Sc		22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn
5	37 Rb	38 Sr	39 Y		40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd
6	55 Cs	56 Ba	57 La	*	72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg
7	87 Fr	88 Ra	89 Ac	**	104 Rf	105 Db	106 Sg	107 Bh	108 Hs	109 Mt	110 Ds	111 Rg	112 Cn
				*	58 Ce	59 Pr	60 Nd	61 Pm	62 Sm	63 Eu	64 Gd	65 Tb	66 Dy
				**	90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf

5) The Sun- M_H=1.00784 Amu 1Amu=1.67*10^-27Kg

M_{he}=4.0026 Amu

6) The Path of Starlight-



Year 1991, Solar Eclipse
island of Príncipe off the west coast of Africa

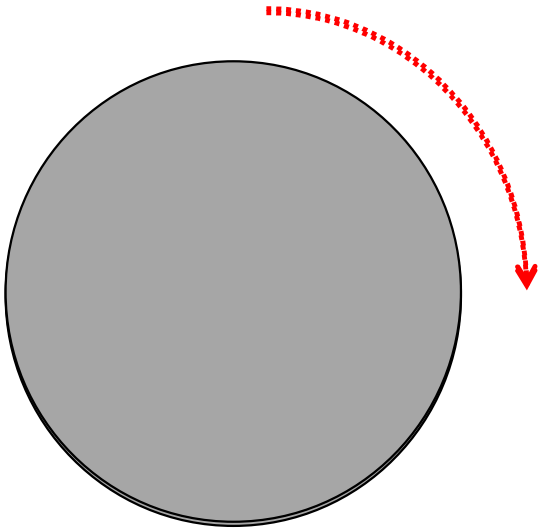
Oh leave the wise our measures to collate
One thing at least is certain, light has weight
One thing is certain and the rest debate
Light rays, when near the sun, do not go straight.

7)GPS- accuracy, the satellites use clocks that are accurate to a few nanoseconds (billionths of a second). Because each satellite is 12,600 miles (20,300 kilometers) above Earth and moves at about 6,000 mph (10,000 km/h), there's a relativistic time dilation that tacks on about 4 microseconds each day. Add in the effects of gravity, and the time dilation effect goes up to about 7 microseconds (millionths of a second). The difference is very real: If no relativistic effects were accounted for, a GPS unit that tells you it's a half mile (0.8 km) to the next gas station would be 5 miles (8 km) off after only one day

Trying to Break the Cosmic Limit & Paradox

As strange as it may seem the cosmic limit or the speed limit can actually be broken.

Let's say we have a laser (Light amplification by stimulated emission of radiation) which is intense enough to reach the surface of moon. The light (say photon) will travel at the max. speed which is 'c'. Let us fix the laser in a spinning device which can produce 100 revolutions per second. It means the Laser will cover the circumference of the moon 3x the speed of light. Thus, we have achieved a speed that can reach things faster than the ultimate speed limit, or is it?
Well, in reality no photon is actually travelling at speed higher than 'c'. It is the dot which propagates in the direction perpendicular to the photons.



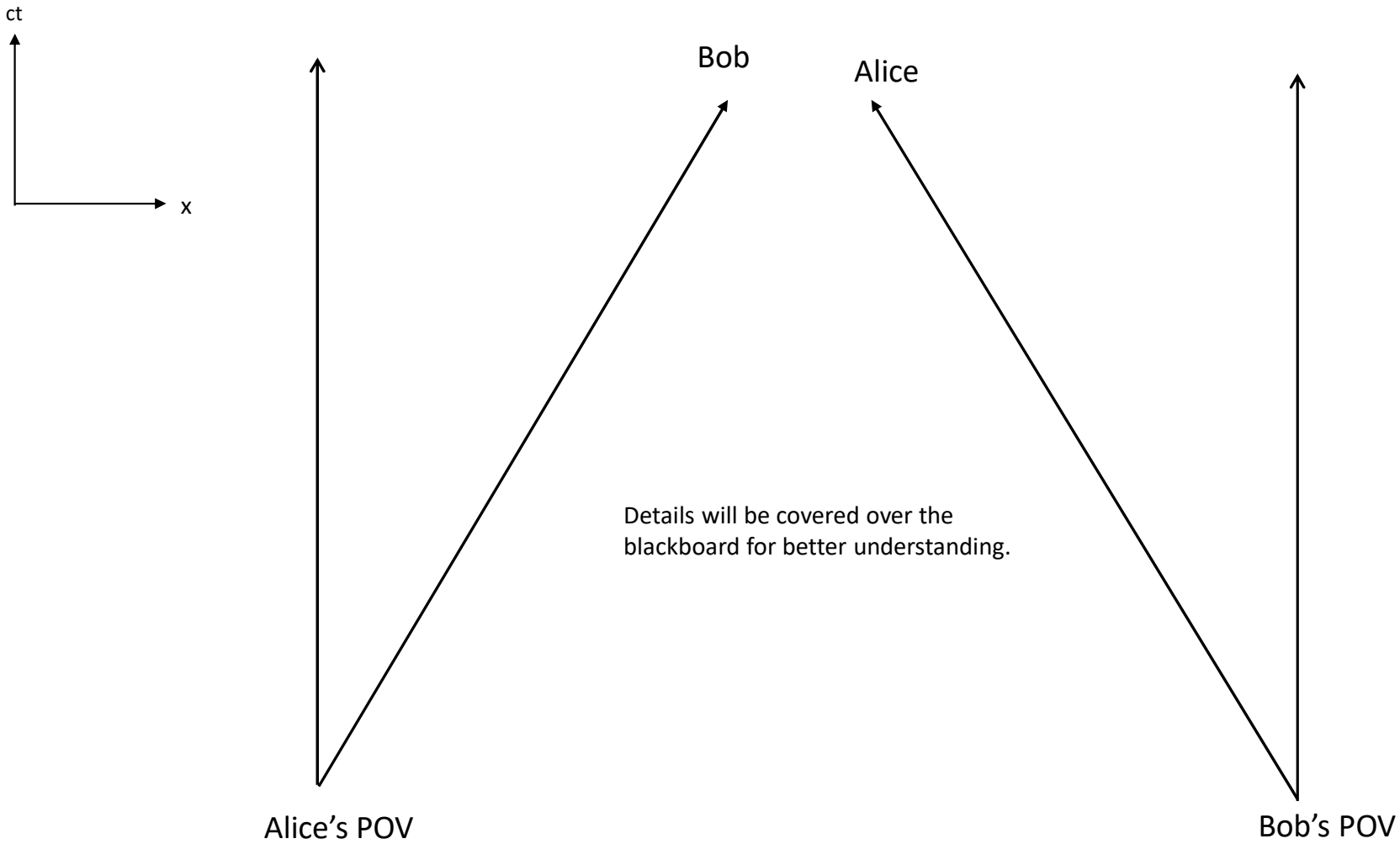
Which means that, there's no absolute object that is travelling. The dot doesn't carry any information with it and it literally cannot cause anything, unlike the photon.

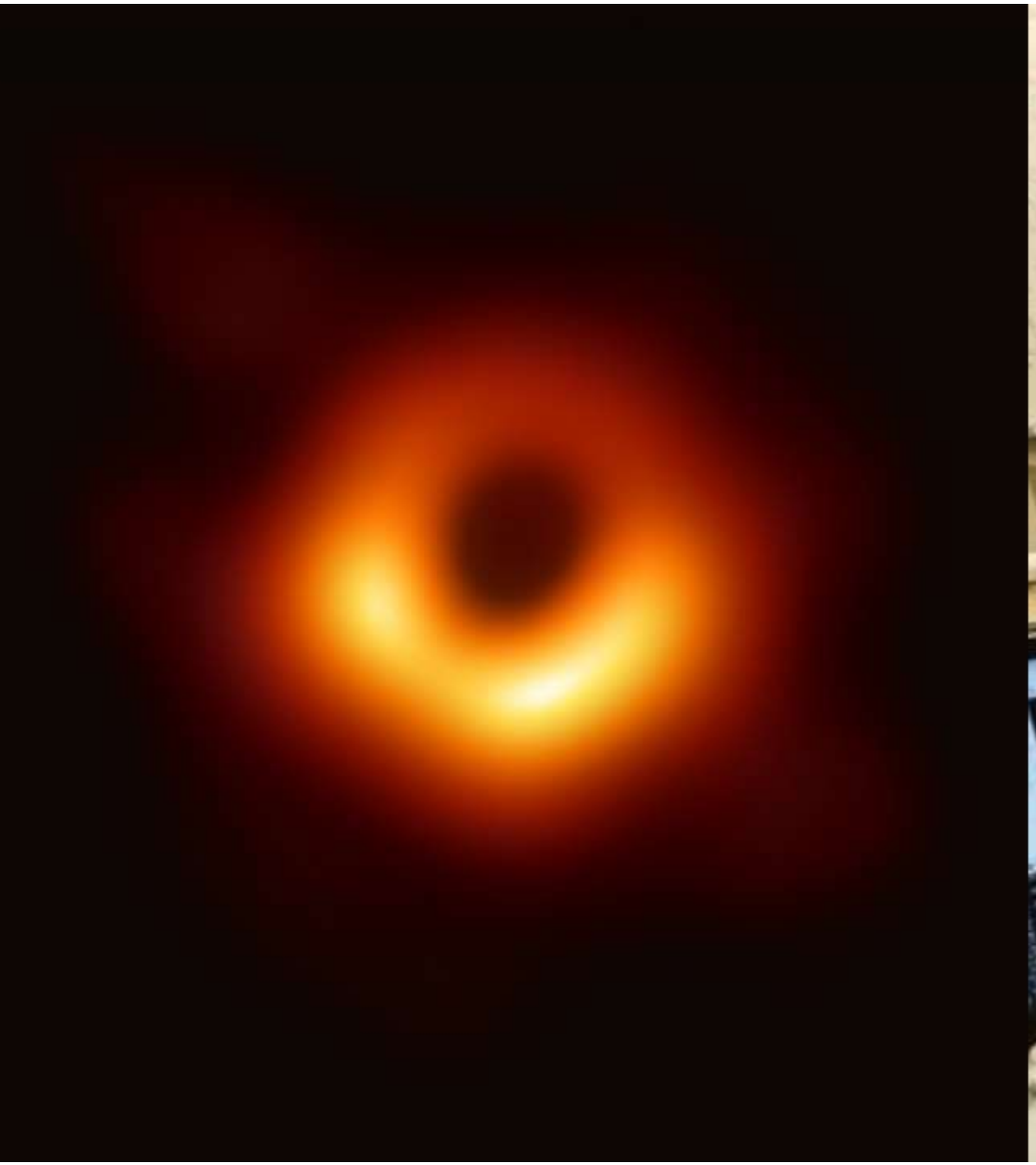
Photon is just an excitation in the E-M field.

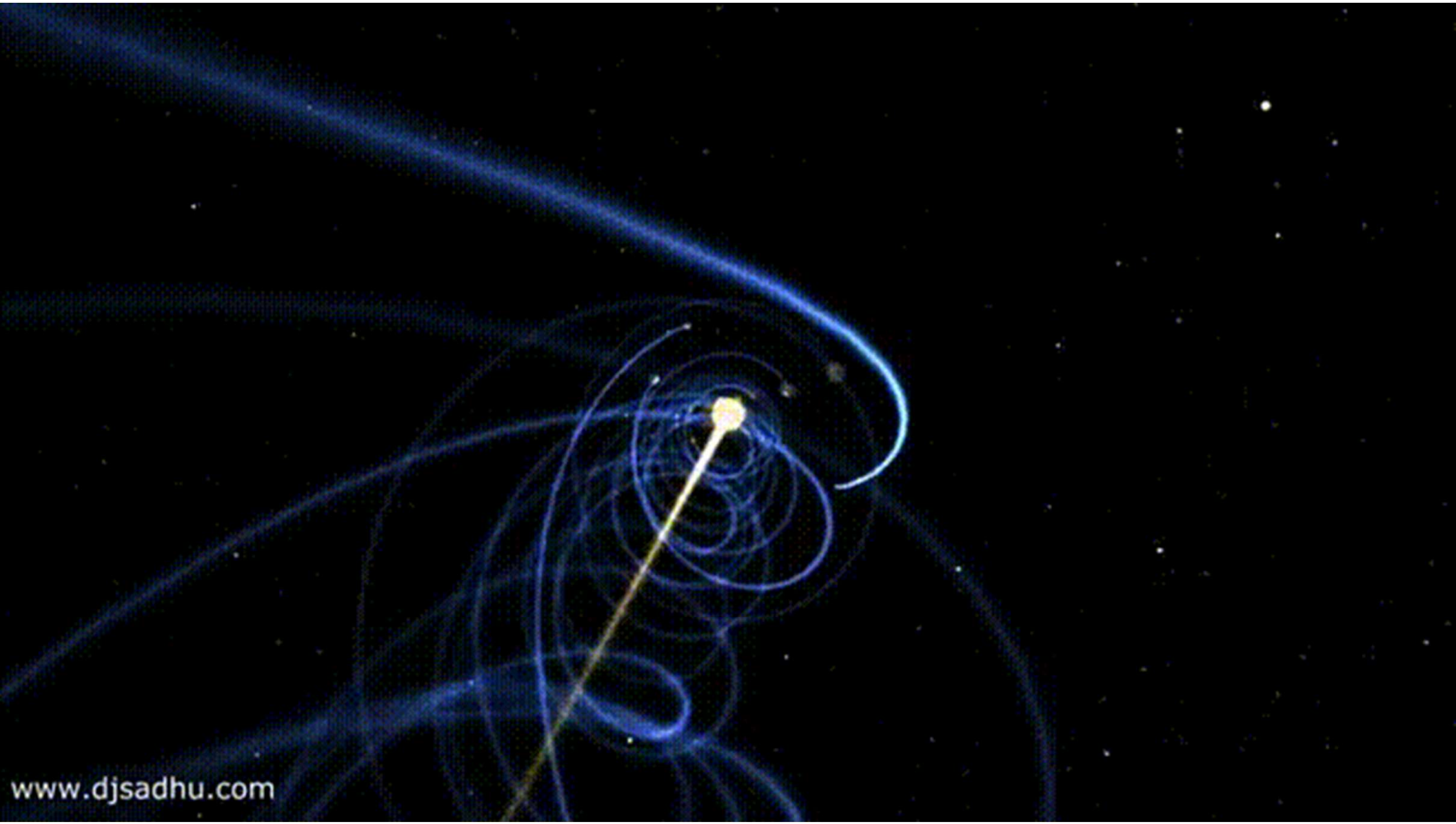
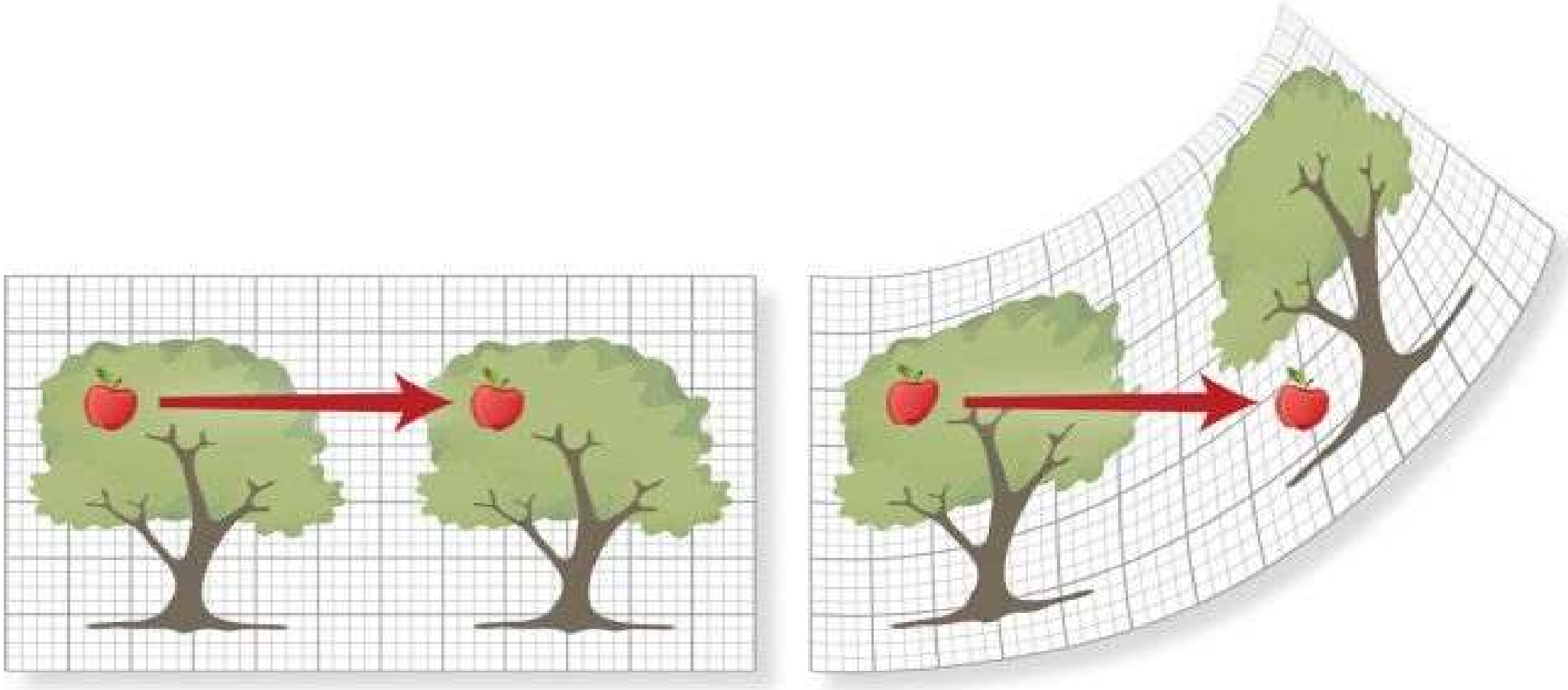
Speed of Causality = Speed of Force Field

0.87 c

0.87 c


ct
x
Bob
Alice
Details will be covered over the
blackboard for better understanding.
Alice's POV
Bob's POV

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